Vectors

Thursday, March 05, 2026 11:30 AM

- A vector is an ordered list of numbers. It can be represented vertically or horizontally.

Ex: $\begin{bmatrix} -1.1 \\ 0.0 \\ 3.6 \\ -7.2 \end{bmatrix}$ or $\begin{pmatrix} -1.1 \\ 0.0 \\ 3.6 \\ -7.2 \end{pmatrix}$ or $(-1.1, 0, 3.6, -7.2)$

elements, entries, or coefficients

In this example, a_1 would be the 1st vector's element ($a_1 = -1.1$), a_2 would be the 2nd ($a_2 = 0$), and so on. Thus, a_n would be the nth element in an n-vector (vector w/ n elements).

- Given two vectors, a and b. If they have the same size (number of elements), we can write $a = b$ if every $a_i = b_i$. This is called overload.
- This is the mathematical notation - which is different from computational notation (it varies according to each programming language).
- Block vectors:

Ex: given two vectors s and t. Considering the vector v is the stacked vector or concatenation of s and t, we have:

$s = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ (size 2) and $t = \begin{bmatrix} 0 \\ 2 \\ 2 \end{bmatrix}$ (size 3) $\therefore v = \begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 2 \\ 2 \end{bmatrix}$ (size 2+3=5)

- Unit vector: one element is 1 and all the others are 0s. It's denoted e_i, where the ith entry is the 1.

Ex: $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ or $e_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ or $e_5 = (0,0,0,0,1)$

- 0_n is the notation for the n-vector where all entries are 0s.
- 1_n is the notation for the n-vector where all entries are 1s.
- We say that a vector is sparse when many of its entries are 0.
  - ↳ Sparse vectors can be stored and manipulated efficiently on a computer.
  - ↳ $nnz(x)$ is the number of entries that are nonzero
  - ↳ Zero vectors and unit vectors are examples of sparse vectors.
- Slicing of vectors.

Ex: $a = (1,2,3,4) \rightarrow a_{\overset{t}{2}:\overset{s}{3}} = (2,3)$

> The size of the sliced vector is $s - t + 1$

Vector Addition

- Two vectors can be added together when they are the same size.

Ex: $a = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, b = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \therefore a + b = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$

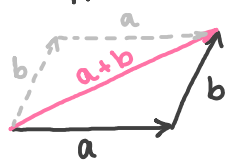
> The size of the resulting vector is the same as the added vectors'.

> Subtraction is similar

- Properties:
  - ↳ Commutative → $a + b = b + a$
  - ↳ Associative → $(a+b)+c = a+(b+c) = a+b+c$
  - ↳ $a + 0 = 0 + a = 0$ → 0 is the same size as a
  - ↳ $a - a = 0$ → 0 is the same size as a

> Remember that, even though these properties look obvious, we are operating with vectors – not num

- This also applies to displacements.



It doesn't matter if we "walk" through a first and then b, or through b firs and then a, because in the end, we walked the same distance and reach the same point.

Scalar-vector Multiplication

- We can multiply a vector a for a scalar β to obtain the vector βa as a result.

Ex. $\beta = 3$, $a = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \therefore \beta a = 3 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$

- Properties:
  - ↳ Associative → $(\beta\gamma)a = \beta(\gamma a) = \beta\gamma a$
  - ↳ Left distributive → $(\beta + \gamma)a = \beta a + \gamma a$
  - ↳ Right distributive → $\beta(a+b) = \beta a + \beta b$
  - ↳ Commutative → $\alpha a = a\alpha$

Linear Combinations

- For vectors $a_1, \dots, a_m$ and scalars $\beta_1, \dots, \beta_m$, $\beta_1 a_1 + \dots + \beta_m a_m$ is a linear combination of the vectors.
- Some linear combinations have special names:
  - ↳ When $\beta_i \geq 0$ and $\sum \beta_i = 1$, we call it weighted average.
- A simple identity: for any n-vector b, $b = b_1 e_1 + \dots + b_n e_n$

Ex. $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

- Intuition: in a linear combination, we are basically combining vectors into a single one.